DATA 3421 – Final Project Proposal

Heart Attack Risk Prediction

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Introduction

Heart disease and specifically heart attacks are problems that can be found in all developing countries. These major health phenomena can affect a vast range of individuals, regardless of age, gender, or physical health. In fact, the risk of both heart attacks and disease has been on the rise. An article by Km & Zoobi (2025) highlights this increased likelihood of heart attacks, particularly in younger individuals, noting that "...the incidence of heart attacks in individuals under 40 has risen by 2% annually until 2016...". The article further indicates that the COVID-19 pandemic exacerbated this issue, contributing to the more recent increased risk of heart attacks (Km & Zoobi, 2025). This growing concern is important as it indicates that a significant number of people are not making the best choices regarding their health, and this trend has not yet shown a substantial decrease.

It is essential to have a prevention strategy that promotes healthier lifestyle choices, encourages regular screenings, and supports the management of pre-existing conditions. Identifying early indicators of heart attack risk can play a critical role in creating such prevention efforts. To help mitigate the risk of heart attacks, we wish to establish early indicators for this risk, whether it be family history, age, BMI, or other factors.

We chose to analyze a dataset looking at 24 indicators across 8763 different people to establish different trends and relationships between these indicators and heart attack risk.

Proposed Data Mining Plan

This is a classification problem, deciding between the presence or absence of heart attack risk. There are numerous categorical and numerical variables to analyze with opportunities for feature engineering (blood pressure measures numerical values but is initially given as a string). Making prudent choices for standardizing/label encoding these variables will help analyze the data. Since this is a healthcare related study, accuracy, precision, and recall will be the top priority as a false negative would prove disastrous. Not only can we classify the presence of a heart attack, but we can also determine the relationships between different indicators (i.e., diet and heart rate). We can demonstrate which indicators have the most significant impact on the heart and which are not likely to indicate risk. These data can aid in identifying people at most risk of a heart attack and allow for early prevention.

We will go through the standard preprocessing steps: one-hot encoding categorical features and scaling numerical features. The methods for scaling will depend on the distribution of numerical data. For normal features, normalization will be used while techniques such as log transforms may be used for skewed or non-normal data. These techniques will allow us to implement a variety of machine learning models to answer the question of heart attack risk. We have chosen to build several classification models, including: decision tree, random forest, k-nearest neighbors, and logistic regression. Since this problem involves outcomes related to healthcare where accuracy can be a matter of life and death, we will evaluate all these models against standard machine learning metrics, such as accuracy, precision, and recall.

References

Km, D., & Zoobi, T. (2025). The Rising Incidence of Sudden Heart Attacks: Causes, Risk Factors, and Prevention Strategies. *TMP Universal Journal of Medical Research and Surgical Techniques*, *1*(1). https://doi.org/10.69557/kf513732